

(MATTHEW) YIZHENG HE

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Education

McGill University, Montreal, Canada

2023 – 2027 (expected)

B.A. & Sc. Honours Cognitive Science (Computer Science Focus), Major Mathematics (in progress)

CGPA: 4.0/4.0 (Top 5%)

Research Experience

Masset Lab *Research Intern*

↪ Prof. Paul Masset

2024–Present

McGill University, Mila–Quebec AI Institute

Projects involved:

- **Simultaneous detection and estimation in olfactory sensing (Sep. 2024–Present)**
(Recurrent neural networks; Constrained optimization; Bayesian inference; Mirror-Langevin descent)
Designed, coded, and ran scaling experiments on HPC clusters; refined model variants to improve performance.
First author on the accepted abstract and co-first author on the associated preprint (under review at PNAS).
- **Population structure of reward-induced remapping in hippocampal CA1 (Apr. 2025–Present)**
(Representation similarity analysis (RSA); Tensor decomposition; Dimensionality reduction techniques)
Built data pipelines for preprocessed data; performed exploratory data analysis and developed metrics to quantify neural population structure and reward-related remapping. Made figures for the accepted COSYNE abstract.
- **Piecewise deterministic spiking samplers (Nov. 2025–Present)**
(Piecewise-deterministic Markov processes; MCMC sampling algorithms; Spiking neural networks)
Participating in both simulation-based and theoretical investigation.

Research Interests

Computational & Theoretical Neuroscience; Machine Learning Theory

Preprint

Chen Jiang*, **Matthew Y. He***, Venkatesh N. Murthy, Cengiz Pehlevan, Jacob A. Zavatone-Veth[†], and Paul Masset[†] (2025). “Simultaneous detection and estimation in olfactory sensing”. In: *bioRxiv*. (Under review at PNAS). doi: [10.1101/2025.11.01.686013](https://doi.org/10.1101/2025.11.01.686013).

* Equal contribution. [†] Co-supervision.

Conference Abstract

Matthew He, Chen Jiang, Venkatesh Murthy, Cengiz Pehlevan, Jacob Zavatone-Veth, and Paul Masset (2025). “Simultaneous detection and mapping in the olfactory bulb”. In: *Computational and Systems Neuroscience (COSYNE)*.

Chen Jiang, Adel Halawa, **Matthew He**, and Paul Masset (2026). “Population structure of reward-induced remapping in the hippocampal CA1”. In: *Computational and Systems Neuroscience (COSYNE)*.

Awards & Distinctions

Faculty of Science Scholarship

2025-2026

Established in 1992 by the University to provide awards based on academic achievement to students in the top 5% of the Faculty of Science.

Dean's Honour List

2023-2025

Top 10% of students in the Faculty of Science based on the combined GPA for the fall and winter terms.

Science Undergraduate Research Awards

2025

\$8950 CAD for 15 weeks of full-time research in the summer of 2025.

SUS Ambassador Fund

2025

\$750 CAD for traveling to present at Computational and Systems Neuroscience (COSYNE) 2025 conference.

Tomlinson Undergraduate Award

2025

Mentoring students in a molecular and cellular biology course.

Conference & Workshop Attendance

Mathematical Foundation for Data Science *Stochastic Optimization; High-dimensional Statistics* **May–June 2025**
↪ Centre de Recherches Mathématiques, Montreal, Canada

Computational and Systems Neuroscience (COSYNE) 2025 *Conference & Workshop* **April 2025**

SynBio Collective Wet Lab Workshop Series *Theme: Expressing recombinant genes in bacteria* **January–April 2024**
↪ McGill iGEM Team

Selected Coursework

Graduate-level	Undergraduate-level (selected)
Stochastic Processes (MATH547) [†]	Nonlinear Optimization (MATH378)
Computational Neuroscience (NEUR503) [†]	Honours Applied Linear Algebra (MATH247)
Numerical Analysis (MATH578) [†]	Honours Nonlinear Dynamics (MATH376)
Expected by Summer 2026	Honours Algorithms & Data Structures (COMP252)
Theory of Machine Learning (MATH562) [†]	Probability & Statistics (MATH323,324)
Honours Convex Optimization (MATH563) [†]	Discrete Structures (MATH240)
Representation Learning (IFT6135) [†]	Cognition (PSYC213)
Honours Analysis 2 (MATH255)	

[†] Graduate level.

Skills

Scientific Computing	Python (PyTorch, JAX), MATLAB, Linux, HPC clusters (SLURM)
Mathematical Areas (beyond coursework)	Bayesian inference, sampling methods, tensor decomposition, physics-informed neural networks (PINNs)
Neuroscience Knowledge	Hippocampus, olfactory bulb, midbrain dopaminergic neurons; general systems neuroscience knowledge learned through courses